

Volume I of The Beholder Project

BEYOND PARAMETER SCALING

A First-Principles Theory of
Emergent Machine Intelligence

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Beyond Parameter Scaling
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This work is presented as a conceptual manuscript.

How to Read This Book

- Do not treat biological analogies as proofs. They are generators of hypotheses.
- Do not treat current LLM capability as evidence of consciousness.
- Do not treat multi-agent evolution as guaranteed AGI.
- Do not treat parameter scaling as obsolete. It is one important axis in a larger proposed scaling space.
- Do not treat the author's personal timeline as a substitute for literature.

Executive Convergence

The strongest defensible conclusion from the full line of reasoning is not that parameter scaling is useless, nor that the Omega concept is already AGI. It is that machine intelligence is better treated as a multi-dimensional system in which model capacity is one variable among several interacting variables: knowledge, exploration, memory, perception, abstraction, inference-time computation, multi-agent interaction, evolutionary selection, analogy, hypothesis generation, and independent evaluation.

Recent literature validates several components independently. LLMs exhibit analogical reasoning; open-ended environments produce generalizable behaviors; evolutionary LLM-agent systems discover and improve algorithms; automated evaluators separate generation from verification. What remains unestablished is whether combining these components around a sufficiently capable neural learner is enough to produce AGI.

The Core Propositions

P1 — Intelligence is not one-dimensional

Status: Established direction

Parameter count, data and compute matter, but capability also depends on training, inference-time computation, tools, memory, environment and task structure. Scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) confirmed the power of parameter scaling while suggesting its limits. Inference-time computation (Snell et al., ICLR 2025) has emerged as a distinct scaling axis.

P2 — Exploration and validation should be separable

Status: Research hypothesis

AlphaEvolve (DeepMind, 2025) separates LLM generation from automated evaluation. FunSearch (2024) used the same separation for mathematical discovery. Dung's argumentation framework (1995) and abstract dialectical frameworks (Brewka et al., 2013) provide formal structures for independent validation.

P3 — Analogy is a mechanism for novelty

Status: Strongly supported

Webb, Holyoak & Lu (Nature Human Behaviour, 2023) demonstrated emergent analogical reasoning in LLMs. Shen et al. (2026) reported

90–173% increases in solution diversity from explicit cross-domain analogy. LLMs show bias toward near analogies (arXiv:2608.03233), and strongest results come from human-LLM collaboration.

P4 — Contradiction can be retained as a productive constraint

Status: Research hypothesis — formally grounded

Grounded in: (1) Paraconsistent logic (Priest, 1979) allowing contradictions without collapse. (2) AGMp belief revision (Ribeiro et al., 2013) extending AGM theory. (3) Abstract dialectical frameworks (Brewka et al., 2013) modeling competing arguments. The five-step algorithm: preservation, structural analysis, synthesis search, validation, revision.

P5 — Population-level evolution can improve solutions

Status: Established direction

XLand (DeepMind, 2021), AlphaEvolve (2025), Chen et al. (2025, co-evolution), Hu et al. (2026, EvoMAS), CORAL (Qu et al., 2026, autonomous evolution). OpenEvolve (2025) provides reproducible open-source implementation.

P6 — Horizontal and vertical scaling are distinct

Status: Research hypothesis

NeurIPS 2026 and ICLR 2026 workshops reflect growing field recognition. No study has yet held the base model fixed while independently varying abstraction depth and population size.

P7 — Perception and embodiment change intelligence

Status: Decomposed hypothesis

P7a: Multimodal input increases accessible information (established). P7b: Sensorimotor interaction produces different representations (strongly supported). P7c: Embodied feedback may be necessary for valuation (open). P7d: Consciousness required for AGI (philosophical, excluded).

P8 — Capability and governance can be separated

Status: Governance hypothesis

The EU AI Act (2024) implements this approach legally. SSRN (2025) extends it to internal architectural governability.

P9 — Creator/creation autonomy is philosophical

Status: Philosophical framing

Valuable as a lens on creation, freedom, and participation. Not empirical evidence for consciousness. Standard references: Searle (1980), Chalmers (1996).

Omega — Formal Research Architecture

Omega is a research architecture whose central loop is: Knowledge → Open-ended exploration → Candidate generation → Analogy → Contradiction-preserving synthesis → Hypothesis → Independent validation → Persistent knowledge → Next cycle.

> *Let K_t be knowledge state, E_t the environment, G_t the candidate set, S the synthesis operator, V the validator:*
>
> $G_t = Generate(K_t, E_t); H_t = S(G_t); r_t = V(H_t); K_{t+1} = K_t \cup Valid(H_t)$

Formal Grounding for P4

The system maintains K_t as a paraconsistent belief set under AGMp revision. Competing hypotheses are retained with designated truth value b. Structural analysis extracts the symmetric difference of assumptions. Synthesis search finds H' subsuming both competitors under context restrictions. If validated, classical AGM revision replaces competitors with synthesis; if not, the contradiction is preserved.

Falsifiable Experimental Program

Use the same base LLM and matched compute. Compare C0 (single agent) through C9 (open-ended environment + all mechanisms). Measure literature novelty, solution diversity, validity, reproducibility, transfer, compute efficiency, and rate of verified discoveries.

Condition	Description
C0	Single agent, no external memory
C1	Single agent + retrieval
C2	Single agent + persistent memory
C3	Multiple agents + communication
C4	Multiple agents + independent validation
C5	Multiple agents + competition
C6	Population + mutation/selection
C7	Population + tool creation
C8	Population + multimodal perception
C9	Full ecology

Key control: Compare monolithic model against smaller model in full ecology under matched compute. $\Delta I_{ecology} = I_{ecology}(B) - I_{monolith}(B)$.

Limits, Objections and Failure Modes

- The biology analogy may be misleading.

- More agents can create coordination overhead.
- Evolution can optimize proxies and produce pathological strategies.
- Validation can itself be wrong or gameable.
- There may be hard limits in the fixed core model that ecology cannot overcome.

The most important objection: perhaps the LLM core is already a highly specialized artifact whose limits cannot be overcome by surrounding it with more copies. The answer is the control experiment.

Let It Be Broken

A useful scientific idea should not be protected from criticism. It should be exposed to it. Perhaps intelligence is mostly parameter scale. Perhaps the combination will produce a capability regime nobody currently understands. All of these possibilities remain open.

The next step may not be another larger box. It may be a world. That is the invitation of this book: Do not believe it. Build it. Measure it. Break it. Improve it.

Appendix A — Mathematical Core

A.1 Resource-normalized capability

I(B) = performance under resource budget B.

A.2 Scaling elasticity

ε_x = (ΔI/I) / (Δx/x).

A.3 Interaction synergy

S_{xy} = I(x,y) – I(x,0) – I(0,y) + I(0,0).

A.4 Diversity

D(P) = H(Strategies(P)).

A.5 Validation reliability

R_V = P(Validator correct | c).

A.6 Open-endedness

O = Novelty × Sustainability × Unpredictability.

Appendix B — Terminology and Claims Discipline

Emergence

System-level property from interactions. Does not imply consciousness.

Intelligence

Adaptive competence across tasks. Deliberately operational.

Consciousness

Subjective experience. No claim current AI possesses it.

Evolution

Iterative variation, selection, inheritance. Not mere repeated prompting.

Ecology

The interacting system of agents, tools, knowledge, memory, evaluators and tasks.

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Intelligence is not a single number. It is not the count of parameters in a neural network, the volume of training data, or the FLOPS of a compute cluster. These matter — but they are one axis in a space that has at least nine.

Beyond Parameter Scaling proposes a first-principles research framework arguing that machine intelligence is an emergent property of a computational ecology: a system of knowledge, exploration, analogy, contradiction, multi-agent interaction, evolutionary selection, and independent validation operating around a neural learner.

Drawing on paraconsistent logic, argumentation theory, evolutionary computation, and the latest empirical results from frontier AI research, this book formulates nine testable propositions, a formal research architecture (Omega), and a falsifiable experimental program that can be run today.

The author's invitation is simple: Do not believe it. Build it. Measure it. Break it. Improve it.

About the Author

Niloy Das is a production software engineer and independent researcher whose work spans machine intelligence, AI-assisted filmmaking, and the philosophy of creation. He is the creator of the Beyond Parameter Scaling framework, the GODSHAND AI filmmaking toolkit, and The Beholder Project — a three-volume exploration of intelligence, creativity, and the sacred. He lives and works at the intersection of first-principles engineering and contemplative inquiry.